



C23-CM/CIOT-302

23189

BOARD DIPLOMA EXAMINATION, (C-23)
MARCH/APRIL—2025
DCME – THIRD SEMESTER EXAMINATION
DIGITAL ELECTRONICS

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. List various number systems used in digital computer along with their base values.
- 2. Convert $(22.125)_{10}$ to an equivalent binary number.
- 3. Give Sum of Products (SOP) Boolean expression for the below truth table :

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

4. What are the advantages and disadvantages of parallel adder?
5. Draw NAND latch with truth table.
6. What is meant by edge triggering. Level triggering in Flip-flops?
7. Differentiate between synchronous and asynchronous counters.
8. List different types of data transfer in registers.
9. Draw the logic symbol of 4×1 Multiplexer.
10. Define Programmable Logic Device.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. Subtract the following numbers using 2's complement method :
(a) 110101-101001 and (b) 1010-11
12. Implement AND, OR, NOT, EX-OR gates using only NAND gates.
13. Explain the operation of Digital Comparator circuit for two 4-bit words.
14. Explain the working of D Flip-flop with block diagram, waveforms and truth table.
15. Explain JK Flip-flop with circuit diagram, waveforms and truth table.
16. Draw and explain Modulo-16 Synchronous counter circuit diagram with waveforms and truth table.
17. Explain the working of Universal Shift Register.
18. Draw and explain the operation of BCD to Decimal decoder circuit diagram with truth table.